Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 10 · SLO 2 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 10 : SLO â€“ 2 | Implementation of Queue Using Linked List â€” Code

I. Node Structure

• struct node { int data; struct node *next; };
• struct queue { struct node *front, *rear; };

II. Enqueue Operation

• void enqueue(struct queue *q, int val) {
• struct node *ptr = (struct node *)malloc(sizeof(struct node));
• ptr->data = val;
• ptr->next = NULL;
• if (q->front == NULL) { q->front = q->rear = ptr; return; }
• q->rear->next = ptr;
• q->rear = ptr;
• }

III. Dequeue Operation

• int dequeue(struct queue *q) {
• if (q->front == NULL) { printf("UNDERFLOW\n"); return -1; }
• struct node *ptr = q->front;
• int val = ptr->data;
• q->front = q->front->next;
• if (q->front == NULL) q->rear = NULL;
• free(ptr);
• return val;
• }